Menu Intelligence: Structured Menu Extraction and Categorization

Andrey Philippov

April 2026

Abstract

Menu Intelligence is an end-to-end service for converting pasted menu text, PDF menus, and menu images into a structured JSON menu representation. The project is framed as an NLP pipeline with three main layers: document text extraction, item-level category classification, and deterministic structured field extraction for names, prices, and sizes. The final course dataset is `items.v2.jsonl`, a derived item-level dataset with 432 annotated menu items from 12 public restaurant or hotel dining sources, using source-level train/validation/test splits. On the held-out `items.v2` test split, the strongest model is a TF-IDF word/character n-gram feature union with Logistic Regression, which reaches 0.7361 accuracy and 0.7254 Macro-F1. This classifier is the current backend category model, while menu item field extraction remains deterministic. The repository is available at `https://github.com/philippovdev/nlp-menu-intelligence`.

1 Introduction

Restaurant menus are still frequently published as free-form text, PDFs, or images intended for human reading rather than structured reuse. This creates friction for digital menu management, catalog maintenance, search, delivery integration, and downstream analytics. The practical goal of this project is to build a web service that accepts raw menu content and returns structured menu items that can be reviewed and edited in a browser.

From an NLP perspective, the task is not a single model problem. It is a pipeline that begins with raw document text, continues through line splitting and menu-item detection, predicts a fixed category label for each candidate item, and extracts structured fields such as item name, price, and size. This report therefore treats the system as a composition of measurable tasks rather than as a monolithic application.

Contributions. This project makes four concrete contributions. First, it implements a working menu-structuring service with a deployed frontend and FastAPI backend. Second, it defines a stable item-level JSONL schema and two public dataset slices, culminating in the training-ready `items.v2.jsonl` release. Third, it measures heuristic, classical, and transformer classification baselines on a fixed split. Fourth, it integrates the best measured classifier into the real backend parsing pipeline while preserving deterministic field extraction and the existing API contract.

1.1 Team

This is a solo project. All work was completed by one contributor: problem definition, dataset curation, annotation design, baseline experiments, model training, backend and frontend implementation, deployment, and report preparation.

2 Related Work

The closest published work to Menu Intelligence does not come from one single benchmark. Instead, the relevant literature splits across three neighboring areas: short text and product-title categorization, food-

Current system pipeline. Text / PDF / image input → text extraction → line splitting and normalization → header, item, and noise detection → TF-IDF word/character union + Logistic Regression category prediction with heuristic fallback → deterministic name, price, and size extraction → structured JSON response and review UI.

Figure 1: Pipeline overview of the current parsing pipeline from document intake to structured JSON output.

domain language understanding, and structured extraction from semi-structured restaurant or document text.

Short text and product categorization work motivates the classical baselines used in this project. fastText shows that linear text models remain strong and efficient on short classification inputs [Joulin et al., 2017]. Product title categorization papers report high accuracy on fixed taxonomies and provide the closest mature analog to menu item classification, even though their domains are e-commerce rather than restaurant menus [Xia et al., 2017, Cevahir and Murakami, 2016].

Food-domain NLP is more fragmented. Prior work includes food-item healthiness classification [Wiegand and Klakow, 2013 recipe-to-dish-name recognition [Hu et al., 2023], and large-scale menu normalization or clustering [Latif et al., 2025]. These papers confirm that food text has domain-specific lexical and compositional structure, but they do not define the same supervised fixed-label classification task used here.

Structured extraction evidence is stronger in adjacent domains than in menus themselves. Restaurant-order extraction from dialogue [Peng et al., 2017] and receipt or form understanding resources such as FUNSD, CORD, and SROIE [Jaume et al., 2019, Park et al., 2019, Huang et al., 2019] are useful analogs for recovering prices and other fields from noisy, semi-structured text. However, they still do not match the full task shape of this project: short menu item text, fixed category labels, deterministic slot extraction, and upstream OCR support in one public benchmark.

3 Model Description

The implemented system is a hybrid pipeline rather than an end-to-end language model. Raw input can arrive as pasted text, embedded-text PDF, or image. PDF files are processed through embedded text extraction first; images are passed through OCR. The extracted text is then normalized and split into candidate lines. Each line is classified as a category header, a menu item, or noise. Menu-item lines are assigned a category label, and deterministic parsing extracts the item name, price list, and size list before the service returns a structured JSON response for review.

The deployed classification stage is now model-backed. The backend ships a scikit-learn TF-IDF word/character feature union with Logistic Regression trained on `items.v2.jsonl`, loads it at application startup, and uses it as the default item-level category predictor. Deterministic extraction remains rule-based. If the classifier artifact is unavailable or model confidence is too weak, the parser falls back to conservative heuristic categorization. This design keeps category prediction learned, field extraction deterministic, and runtime behavior explicit. ONNX remains a future export target rather than the current shipped runtime.

4 Dataset

The main course dataset is an item-level JSONL corpus built for menu classification and deterministic slot evaluation. Each record stores a stable item identifier, source identifier, restaurant identifier, split assignment, language tag, raw item text, one category label, and structured slots for name, prices, and sizes. The active label set contains 12 classes: `salads`, `soups`, `mains`, `pizza`, `pasta`, `burgers`, `sides`, `desserts`, `breakfast`, `drinks_hot`, `drinks_cold`, and `other`.

Two dataset slices are present in the repository. `items.v1.jsonl` is a small first experimental subset used for early classical baselines. `items.v2.jsonl` is the training-ready release used for the final classification comparison and the shipped backend model. The paired manifest stores public source provenance, collection date, source type, license note, and notes, while raw captures remain outside git.

Approach	Task	Dataset	Metric	Reported result	Relevance to this project
fastText [Joulin et al., 2017]	short text classification	AG / DBPedia and related benchmarks	Accuracy	92.5% on AG with bigrams	strong justification for CPU-friendly linear baselines
Xia et al. [Xia et al., 2017]	product title categorization	millions of Japanese titles, 35 classes	Accuracy	>96%	closest catalog-title analog to fixed menu labels
Latif et al. [Latif et al., 2025]	menu normalization and clustering	>700k menu items	similarity / coverage	0.98 intra-cluster similarity; 100% coverage	closest menu-domain paper, but clustering rather than supervised classification
Peng et al. [Peng et al., 2017]	restaurant-domain structured extraction	restaurant conversations to order records	F1 / whole-order accuracy	71.2 F1 / 51.8 whole-order accuracy	domain-near extraction analog, but not menu-line classification
FUNSD / CORD / SROIE [Jaume et al., 2019, Park et al., 2019, Huang et al., 2019]	document information extraction	forms and receipts	benchmark-specific IE metrics	dataset or competition references rather than one shared score	useful analogs for OCR-noisy field extraction

Table 1: Related work summary. All sources are conceptually relevant, but none is a direct public benchmark match for fixed-label menu item categorization with deterministic slot extraction.

The report uses `items.v2` as the fixed evaluation dataset. It contains 432 annotated menu items from 12 public restaurant or hotel dining sources, split by source and restaurant to avoid leakage between similar items. The release is intentionally a source-grounded synthetic expansion rather than a source-faithful raw crawl: public menus are used as provenance anchors and lexical references, but the released item texts are normalized into consistent English forms with normalized `RUB` prices. This keeps the public repository lightweight and reproducible, but it also limits how directly the dataset reflects raw restaurant menu wording.

5 Experiments

5.1 Metrics

The main classification metrics are accuracy and Macro-F1. Macro-F1 is the primary model-selection metric because the project treats all 12 categories as equally important. For the heuristic end-to-end pipeline, the report also tracks exact-match accuracy for price and size extraction on normalized item-level gold

Statistic	Total	Train	Valid	Test
Restaurants	12	8	2	2
Documents	12	8	2	2
Items	432	288	72	72
Avg. tokens per item	8.70	8.70	8.72	8.68
Categories covered	12	12	12	12

Table 2: Dataset statistics for `items.v2`. Class counts are perfectly balanced at 36 items per label overall and 24/6/6 per label across train/valid/test.

annotations.

5.2 Setup

All trained classifiers use the fixed `items.v2` source-level split: 288 training items, 72 validation items, and 72 held-out test items. The classical baselines use raw item text as input and predict one category label per item. The first transformer benchmark fine-tunes `distilbert-base-uncased` as a sequence classifier with maximum sequence length 64, batch size 16 for training, batch size 32 for evaluation, learning rate 2×10^{-5}, weight decay 0.01, warmup ratio 0.1, and early stopping on validation Macro-F1.

The product system itself is broader than the classification experiments: it also supports PDF and image intake, embedded PDF text extraction, OCR for images, deterministic line segmentation, and deterministic slot extraction. However, OCR quality is not the main evaluated contribution of this report. The quantitative comparison therefore focuses on item-level category classification and reports deterministic extraction only for the heuristic pipeline.

To keep end-to-end product quality visible, the repository also includes a small real-world evaluation slice with 12 source-grounded examples: four pasted-text cases, four embedded-text PDF fixtures, and four rendered image fixtures. On this slice, the current deployed backend recovers text, category, price, and size exactly on every example. This set is useful as a lightweight regression check, but it is easier than arbitrary real phone photos and is not used as the main benchmark.

5.3 Methods

Seven methods are compared:

- heuristic end-to-end parser with keyword category rules,
- TF-IDF + Logistic Regression,
- TF-IDF + Linear SVM,
- TF-IDF word/char union + Logistic Regression,
- TF-IDF + sigmoid-calibrated Linear SVM,
- DistilBERT sequence classifier,
- XLM-RoBERTa sequence classifier.

The heuristic system is useful as a product baseline because it runs through the real parser and produces both category and slot outputs. The two TF-IDF baselines provide direct classical comparisons on the fixed split. The DistilBERT and XLM-RoBERTa runs are included as practical transformer benchmarks on the same dataset and split.

Method	Evaluation set	Accuracy	Macro-F1	Notes
Heuristic pipeline	full `items.v2`	0.6181	0.6466	price exact = 1.0000; size exact = 1.0000; not split-comparable
TF-IDF + Logistic Regression	test split	0.7222	0.7009	valid = 0.7361 / 0.7302; strongest plain word-TF-IDF baseline
TF-IDF + Linear SVM	test split	0.7083	0.6930	valid = 0.7500 / 0.7502; stronger validation than plain logistic regression, weaker held-out test
TF-IDF word/char union + Logistic Regression	test split	0.7361	0.7254	valid = 0.7778 / 0.7749; strongest held-out test result and current shipped classifier
TF-IDF + sigmoid-calibrated Linear SVM	test split	0.7083	0.6900	valid = 0.7500 / 0.7431; probability-aware SVM comparison, still below the union model
DistilBERT sequence classifier	test split	0.5556	0.5620	valid = 0.6389 / 0.6248; below both TF-IDF baselines
XLM-RoBERTa sequence classifier	test split	0.5556	0.5462	valid = 0.6250 / 0.6155; multilingual follow-up also below the classical baselines

Table 3: Main measured results. The heuristic pipeline is reported for end-to-end reference but is not directly split-comparable with the trained classification runs because its `items.v2` artifact is measured on the full dataset.

6 Results

Table 3 summarizes the measured results already saved in the repository artifacts. The strongest held-out test result on `items.v2` is TF-IDF word/char union + Logistic Regression with 0.7361 accuracy and 0.7254 Macro-F1. Plain TF-IDF + Logistic Regression reaches 0.7222 accuracy and 0.7009 Macro-F1, while TF-IDF + Linear SVM reaches 0.7083 accuracy and 0.6930 Macro-F1 on the same test split. The probability-aware calibrated Linear SVM remains below the union model, and the two tested transformer baselines also underperform the strongest sparse feature pipeline.

The per-class F1 scores show that the task is not uniformly difficult. The shipped TF-IDF word/char union + Logistic Regression model reaches perfect test F1 on `burgers` and `pizza`, and strong scores on `desserts`, `drinks_cold`, and `salads` (all 0.9091). The hardest classes are `sides` (0.0), `mains` (0.3636), and `other` (0.5263). The dominant confusion is `sides` → `other`, with smaller overlaps between `drinks_cold` and `drinks_hot`, and between broad savory categories such as `mains`, `soups`, and `breakfast`. This pattern suggests that short menu texts with strong lexical anchors are easier than context-dependent categories whose boundaries overlap with other savory dishes.

The transformer results do not support switching away from classical methods yet. DistilBERT reaches 0.5556 test accuracy and 0.5620 test Macro-F1, while the multilingual XLM-RoBERTa follow-up reaches the same 0.5556 test accuracy but only 0.5462 test Macro-F1 on the same fixed split. On this dataset size and label taxonomy, the measured evidence still favors sparse lexical features over the tested transformer configurations. The union model likely benefits from the combination of short menu texts, strong lexical anchors, and character n-grams that help absorb noisy spelling, morphology, transliteration, and OCR-like variation.

6.1 Limitations

The report has four important limitations. First, `items.v2` is a source-grounded synthetic expansion rather than a source-faithful raw crawl, so the released public text is cleaner and more regular than uncontrolled menu captures. Second, the heuristic extraction metrics are measured on normalized item-level annotations and should not be read as a direct OCR robustness result. Third, the first transformer run is only one practical benchmark, not an exhaustive transformer search. Fourth, the current production parser is a hybrid system: category assignment is learned, but name, price, and size extraction remain deterministic by

Input text	Predicted category	Structured output
Caesar with grilled chicken 210 g - 340 RUB	salads	name = Caesar with grilled chicken; price = 340 RUB; size = 210 g
Tomato basil soup 300 ml - 295 RUB	soups	name = Tomato basil soup; price = 295 RUB; size = 300 ml
Double cheeseburger 290 g - 540 RUB	burgers	name = Double cheeseburger; price = 540 RUB; size = 290 g

Table 4: Representative outputs from the current backend parser. The service returns the same item schema for text and file-based parsing.

design.

7 Conclusion

Menu Intelligence delivers a real menu-structuring service and a measurable NLP pipeline behind it. The project now has a stable item-level schema, public dataset releases, a fixed training-ready classification dataset, measured heuristic and learned baselines, and a backend that uses the best measured classifier in production. The strongest held-out result on `items.v2` is TF-IDF word/char union + Logistic Regression at 0.7361 accuracy and 0.7254 Macro-F1, which outperforms the plain TF-IDF baselines, the calibrated Linear SVM comparison, and the tested transformer runs.

The main remaining work is not to redesign the service but to deepen the evaluation. Future steps should focus on richer error analysis, better OCR stress tests, possible transformer follow-up experiments only if they improve over the classical baseline, and ONNX export only after it can preserve the current sparse-model quality. The core report claim is therefore modest but well supported: for this dataset and taxonomy, a hybrid parser with OCR or PDF ingestion upstream, a sparse learned category model, and deterministic field extraction is currently the most practical solution.

References

[Cevahir and Murakami, 2016] Cevahir, A. and Murakami, K. (2016). Large-scale multi-class and hierarchical product categorization for an e-commerce giant. In *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers*, pages 525–535, Osaka, Japan. The COLING 2016 Organizing Committee.

[Hu et al., 2023] Hu, C., Liu, X., and Feng, Y. (2023). Diner: A large realistic dataset for evaluating compositional generalization. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 14938–14947, Singapore. Association for Computational Linguistics.

[Huang et al., 2019] Huang, Z., Chen, K., He, J., Bai, X., Karatzas, D., Lu, S., and Jawahar, C. V. (2019). Icdar2019 competition on scanned receipt ocr and information extraction. In *2019 International Conference on Document Analysis and Recognition (ICDAR)*, pages 1516–1520. IEEE.

[Jaume et al., 2019] Jaume, G., Ekenel, H. K., and Thiran, J.-P. (2019). Funsd: A dataset for form understanding in noisy scanned documents.

[Joulin et al., 2017] Joulin, A., Grave, E., Bojanowski, P., and Mikolov, T. (2017). Bag of tricks for efficient text classification. In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 2, Short Papers*, pages 427–431, Valencia, Spain. Association for Computational Linguistics.

[Latif et al., 2025] Latif, S., Mehmood, A., Turki, S., Ameer, H., Gorban, I., and Fateh, F. (2025). Restaurant menu categorization at scale: LLM-guided hybrid clustering. In *Proceedings of the 18th International*

Natural Language Generation Conference, pages 511–521, Hanoi, Vietnam. Association for Computational Linguistics.

[Park et al., 2019] Park, S., Shin, S., Lee, B., Lee, J., Surh, J., Seo, M., and Lee, H. (2019). Cord: A consolidated receipt dataset for post-ocr parsing. In *NeurIPS 2019 Workshop on Document Intelligence*.

[Peng et al., 2017] Peng, B., Seltzer, M., Ju, Y., Zweig, G., and Wong, K.-F. (2017). May i take your order? a neural model for extracting structured information from conversations. In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 1, Long Papers*, pages 450–459, Valencia, Spain. Association for Computational Linguistics.

[Wiegand and Klakow, 2013] Wiegand, M. and Klakow, D. (2013). Towards contextual healthiness classification of food items - a linguistic approach. In *Proceedings of the Sixth International Joint Conference on Natural Language Processing*, pages 19–27, Nagoya, Japan. Asian Federation of Natural Language Processing.

[Xia et al., 2017] Xia, Y., Levine, A., Das, P., Di Fabbrizio, G., Shinzato, K., and Datta, A. (2017). Large-scale categorization of japanese product titles using neural attention models. In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 2, Short Papers*, pages 663–668, Valencia, Spain. Association for Computational Linguistics.